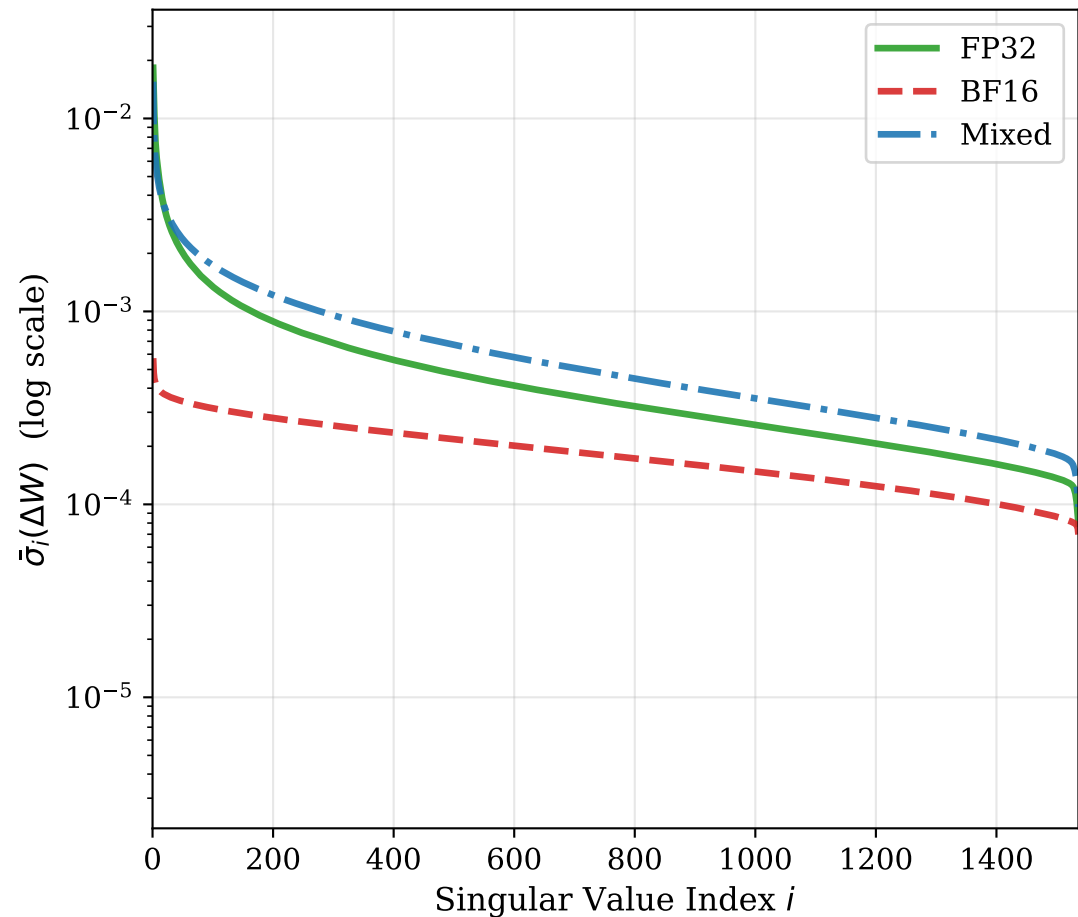


Spectral Decay of ΔW Across All Precision Modes
($\bar{\sigma}_i = \text{mean of } \sigma_i(\Delta W) \text{ across all layers and components}$)

AdamW + GRPO



Muon + GRPO

